Solinas primes and point distributions

2026-10-12 · point-distribution-20261012 · measured observations

None of the 24 primary intervals excludes zero. The study did not resolve a mean difference in these prespecified metrics at the stated simultaneous confidence level. This does not establish equality or exclude other patterns.

What we are measuring

The field is integers modulo p; the curve is $y^2 = x^3 + Ax + B$ over that field. At each x, the right side is either a nonzero square (two points), zero (one point) or a nonsquare (no points). x coverage and point mass are distinct.

With quadratic character χ and $\chi(0) = 0$:

$$w(x) = 1 + \chi(x^3 + Ax + B), \quad N = 1 + \sum_{x \in F_p} w(x).$$

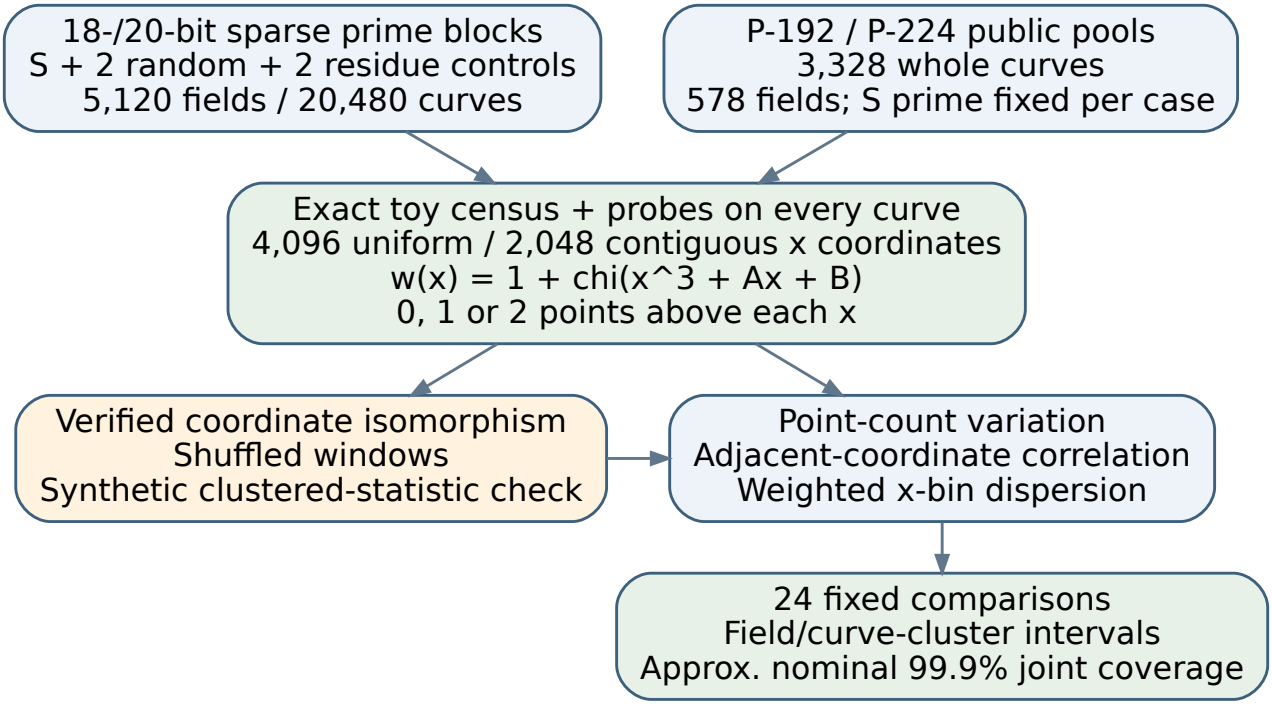
The extra one is infinity. Trace $t = p + 1 - N$ satisfies $|t| \le 2\sqrt{p}$. If Z cubic roots lie in the field, liftable x count is $\frac{N-1+Z}{2}$. Coverage should be close to one half. Local gaps and clustering require separate measurements against matched controls.

For uniformly random nonsingular models, expected coverage is $\frac{1}{2} + \frac{p-1}{2p^2}$ and expected $\frac{t^2}{4p}$ is $\frac{1-\frac{1}{p^2}}{4}$. See THEORY.md for the derivation and exhaustive small-prime checks.

Fixed wider design

20,480 complete curve censuses over 5,120 distinct small primes: 512 blocks each at 18 and 20 bits, four curves per field. Each sparse-prime block has two random and two mod-96 controls, matched within 5% and the same bit size. Sparse means signed nonadjacent binary weight at most six.

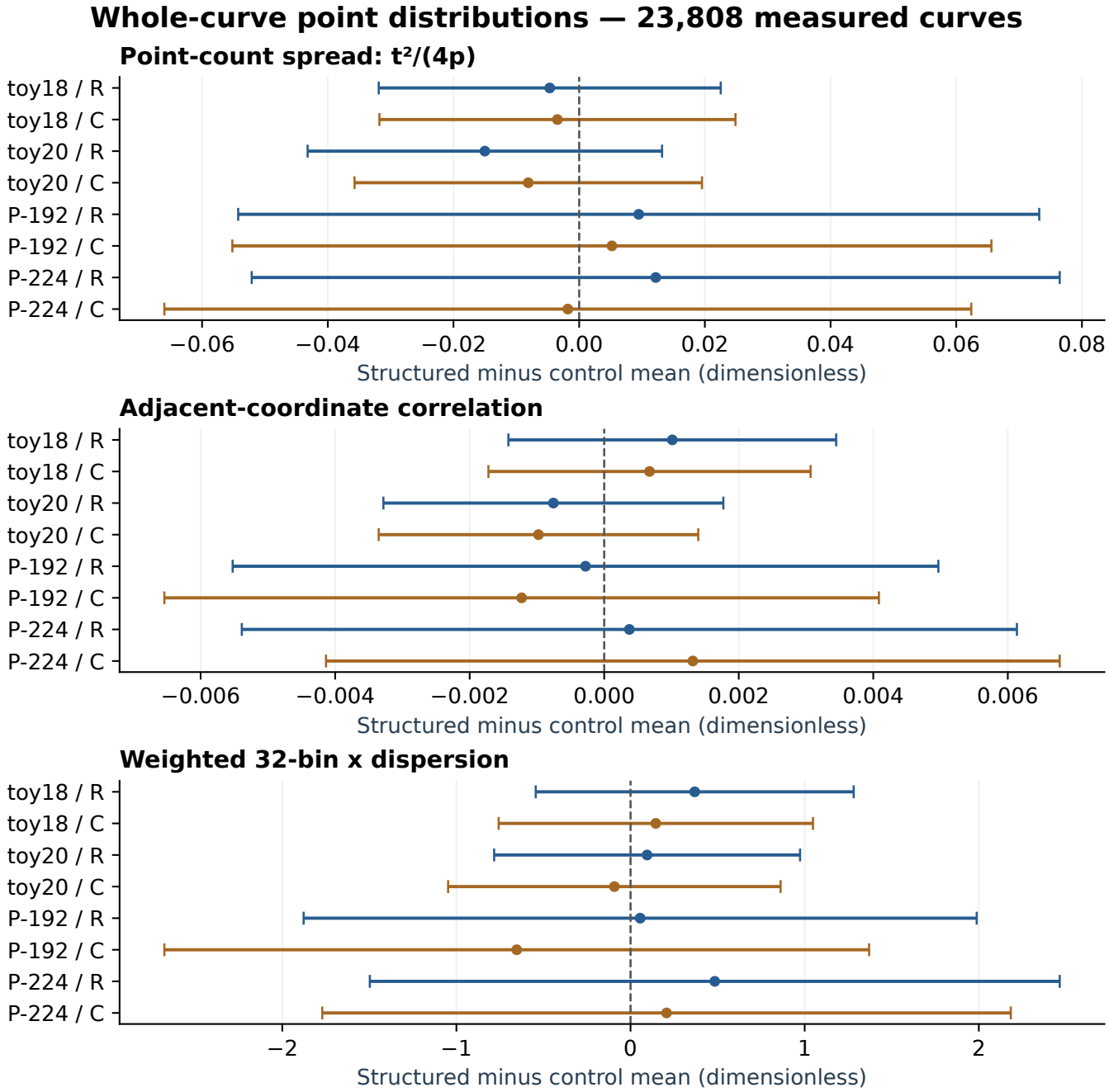
3,328 whole-curve probe cases at P-192/P-224: 512 curves at each fixed structured prime, with 144 random and 144 residue-matched fields per case, four curves per control field. Their existing pools match within 1% and mod 12. Counts are original exact PARI source inputs; x probes are new.



Every curve has 4,096 uniform x probes and 2,048 adjacent coordinates. All outcomes are retained; no outcome-driven stopping. No scalar search.

Primary distribution comparisons

Three metrics: point-count spread $\frac{t^2}{4p}$, adjacent-coordinate liftability correlation and weighted 32-bin x dispersion. Each is compared with R (random) and C (residue-matched). Differences are S minus control.



Bars: approximate Student/Welch intervals, nominal 99.9958333% each; Bonferroni targets nominal 99.9% coverage jointly over 24 comparisons. Toy units: 512 paired prime blocks, all curves in a field kept together. Large-case units: 512 S curve draws at a fixed prime versus 144 control-field means. No exact coverage or power guarantee. Source: analysis.json.

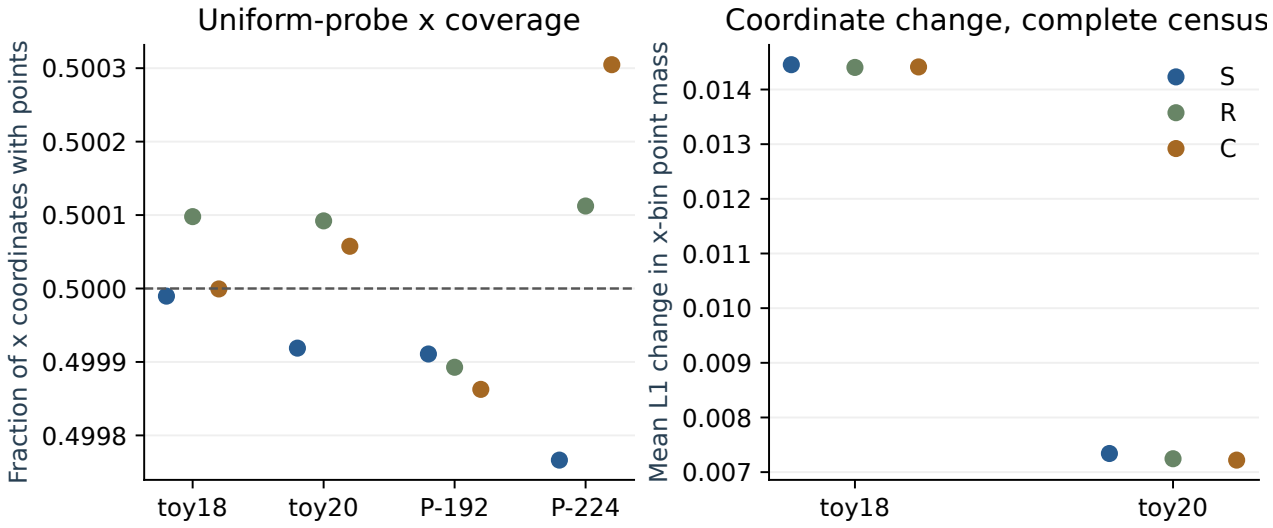
Numerical comparisons and controls

All intervals use the frozen multiplicity correction. Trace², Lag 1 and Bins refer to the three dimensionless metrics above.

Case	Metric	Arm	Difference	Low	High
toy18	Trace²	R	−0.00468	−0.03189	0.02253
toy18	Trace²	C	−0.00346	−0.03179	0.02487
toy18	Lag 1	R	0.00101	−0.00142	0.00345
toy18	Lag 1	C	0.00067	−0.00172	0.00307
toy18	Bins	R	0.36843	−0.54458	1.28143
toy18	Bins	C	0.14520	−0.75731	1.04772
toy20	Trace²	R	−0.01501	−0.04321	0.01319
toy20	Trace²	C	−0.00810	−0.03574	0.01954
toy20	Lag 1	R	−0.00076	−0.00328	0.00177
toy20	Lag 1	C	−0.00098	−0.00335	0.00140
toy20	Bins	R	0.09504	−0.78333	0.97341
toy20	Bins	C	−0.09292	−1.04763	0.86180
P-192	Trace²	R	0.00947	−0.05426	0.07320
P-192	Trace²	C	0.00520	−0.05519	0.06560
P-192	Lag 1	R	−0.00028	−0.00552	0.00497
P-192	Lag 1	C	−0.00123	−0.00654	0.00408
P-192	Bins	R	0.05529	−1.87740	1.98798
P-192	Bins	C	−0.65363	−2.67717	1.36991
P-224	Trace²	R	0.01218	−0.05210	0.07646
P-224	Trace²	C	−0.00181	−0.06601	0.06239
P-224	Lag 1	R	0.00037	−0.00539	0.00614
P-224	Lag 1	C	0.00132	−0.00414	0.00677
P-224	Bins	R	0.48363	−1.49721	2.46447
P-224	Bins	C	0.20673	−1.77026	2.18372

Why coordinate controls matter

The map $(x, y) \to (u^2 x, u^3 y)$ yields $A' = u^4 A$ and $B' = u^6 B$. It preserves the curve order while changing numerical positions. One companion per field is checked; it is not an independent sampling unit. Shuffled windows preserve counts and check the clustering statistics. Synthetic clustered sequences test statistic sensitivity only.



Dots are descriptive means, without inferential intervals. Primary intervals are on page 2. Full toy census bins isolate coordinate changes with N fixed. Companion sampled-window differences also include fresh sampling; they do not alone isolate a coordinate effect. See coordinate-checks.json.

Retained data, checks and limits

Case	Arm	Fields	Curves	Coverage	Lag 1
toy18	S	512	2048	0.499990	−0.00035
toy18	R	1024	4096	0.500098	−0.00136
toy18	C	1024	4096	0.499999	−0.00102
toy20	S	512	2048	0.499919	−0.00123
toy20	R	1024	4096	0.500092	−0.00048
toy20	C	1024	4096	0.500058	−0.00026
P-192	S	1	512	0.499911	−0.00044
P-192	R	144	576	0.499893	−0.00017
P-192	C	144	576	0.499863	0.00078
P-224	S	1	512	0.499766	−0.00023
P-224	R	144	576	0.500112	−0.00060
P-224	C	144	576	0.500305	−0.00154

12 prelaunch tests passed. Post-run checks verify 5,698 field artifacts, 23,808 base curves and 380,928 independently calculated character spot checks, point examples, census identities and checksums. No failed field rows. This is same-session software verification, not an independent scientific review.

Worker CPU: 4649.3 s; run wall time: 1566.7 s, three workers. Raw arrays: 1,981,842,986 bytes; full census masks, zero positions, uniform probe inputs and weights, windows and companion controls retained. Whole-curve URNs hash exact p,A,B; no subgroup generator is selected.

Interpretation limits

The two large structured primes are fixed cases. Their old curve pools are source data, not a fresh sample of a large-prime family. Probe windows are not a whole-field census. Constrained prime matching and approximate intervals limit coverage guarantees. No detected mean difference would not establish equality or exclude higher-order patterns. Numeric clustering alone establishes no group-invariant effect. This run makes no security claim or official hypothesis transition.

Sources and reproducibility

Source commit: `69cdec441c5aec84a18483c32bb7d22d0a387bba`. See report.md for links to protocol, config, inputs.json, environment.json, receipt.json, analysis.json, verification.json and per-field outputs. P-192/P-224 pools: EXP-ECDLP-5184f7 / RUN-ECDLP-d0e449 preparation/curves; exact source hashes are frozen in inputs.json. Derived CSV files list all comparisons and exact whole-curve identities. Full dataset archive checksums are recorded separately. All sources are internal/retrieved local files.

Historical rank, point-supply, attack and ML graphs and frontier files were inventoried. Their supported claims are unchanged; no canonical graph edge is updated. See graph-inventory.json. The editable diagram and SVG figures are kept with this Typst source and PDF.

Proposed next check: lock any newly interesting descriptor and evaluate it on fresh fields and curves. ML may explore this dataset afterward.